

The MILLO Group

Marine Innovation Lab for Leading-edge Oceanography

developing hardware and software to expand the ocean observing network and for the sustainable management of natural resources

Who We Are: We are a group of **Ocean MacGyvers** based in the Department of Ocean and Resources Engineering (ORE) at the University of Hawai'i at Mānoa. We believe good science is as much about the people as the research itself, and we support an exceptional, diverse team committed to making ocean technology more open and accessible.

Vision: We envision a world where **ocean observing is participatory and equitable**, powered by **open, affordable and accessible tools** that accelerate discovery, stewardship and informed decision-making.

Mission: Our mission is to **democratize access to the ocean** by co-designing modular instrumentation and data-processing software that communities can **build, operate and maintain**.

Our Current Work: We design and build **open-source, low-cost tools** that reveal how biological, physical, and chemical processes interact in marine ecosystems and feed directly into science-based solutions.

Tools for Rapid Disaster Response and Restoration including a drifting autosampler to measure coral reef metabolism, a key indicator of reef health; an unmanned surface vehicle with sensors and a water sampler to map water quality in real time after disasters; and a distributed salinity array to detect freshwater runoff following estuary restoration.

Tools for Monitoring Biodiversity and Animal Behavior including underwater acoustic recorders to record reef soundscapes, another key indicator of reef health; a video-audio array to build catalogs linking fish species with their sounds; and statistics-, physics- and AI-based algorithms to detect, classify and track animals in audio and video data, revealing how their behaviors shape ecosystems.

Future Directions include building **island-scale networks of low-cost sensors and samplers** that provide continuous, shared data for reefs, estuaries, and coastal communities. We are also **embedding more computing power and AI directly on our instruments** so they can detect events, trigger sampling, and deliver simple, actionable indicators instead of raw data.

How to Support Our Work:

- **Fund the development of a novel instrument or software package (\$10-50K):** enables us to take a new idea from sketch to field-tested, open-source tool, including prototyping, testing, and documentation.
- **Sponsor a graduate student (\$52.6K):** supports a graduate student researcher who designs, deploys, and maintains our hardware and software and train community operators.

Your Support Helps:

- **Lower the costs of studying and monitoring the ocean** by creating tools that are 10-30× cheaper.
- **Build local capacity** by training students and community partners to operate, maintain, and adapt the technology.
- **Accelerate science and stewardship** by turning open designs and open data into practical, science-based solutions for coastal communities.

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